

Predicting Terrorist Attacks in the US

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Abstract

This project explores the prediction of terrorist attacks in the United States on an annual basis using the Auto Regressive Integrated Moving Average (ARIMA) model and data from the Global Terrorism Database (GTD). Terrorist attacks pose a significant threat to public safety and national security, underscoring the importance of proactive measures. Our objective is to develop a predictive model to estimate the number of terrorist incidents expected between 2025 and 2040, enabling the U.S. to better prepare for and mitigate potential risks. By leveraging the GTD and successfully implementing the ARIMA model, we generated forecasts for future attack trends. The model produced predictions with a percentage error of 63.13%, indicating room for refinement. However, the findings suggest an anticipated increase in terrorist activities in the coming years, highlighting the need for vigilance and strategic planning.

Introduction

Terrorism remains one of the most persistent challenges faced by societies around the world, causing significant disruptions to public safety and economic stability. Predicting such events is a complex task due to their dynamic nature and the multifaceted factors that influence them. Our project aims to develop a predictive model to forecast terrorist attacks by analyzing historical data from the Global Terrorism Database (GTD). The GTD, with over 200,000 records, provides detailed information on attack types, locations, and responsible groups, which served as the foundation for our analysis. The output of our model is a set of predictions that estimate the likelihood of various attack types and locations, which can support decision-making in risk assessment and resource allocation.

The dataset encompasses a range of attack types, including Armed Assaults, Assassinations, Bombings/Explosions, Facility Attacks, Hijackings, Hostage Taking (Barricade Incidents and Kidnappings), and Unarmed Assaults. Each type presents unique patterns and characteristics, offering insights into the strategies used by different terrorist groups. By examining these categories, we identified factors that influence the likelihood and nature of future attacks.

The project addresses several challenges, including the high dimensionality and sparsity of the dataset, biases inherent in historical records, and the need for computational efficiency to handle large-scale data. Initially, we explored several modeling approaches, but none returned meaningful results. Linear regression produced values that were unrealistically small, while Poisson regression generated infinite outputs. Gradient boosting and random forest models returned the same predicted values for each year, failing to capture meaningful trends.

To overcome these issues, we developed a time series model to analyze and predict patterns in terrorist attacks. This involved cleaning and preparing the data, creating features to capture temporal trends, and applying forecasting techniques. To improve the model's accuracy and performance, we fine-tuned parameters and optimized the overall approach.

Unlike prior studies that focus on descriptive analytics, this project emphasizes predictive modeling to uncover actionable patterns and trends. While prior work has provided valuable descriptive insights, few approaches have attempted to forecast future attacks using time series methods. By bridging the gap between data analysis and practical application, this work

contributes to more effective risk assessment and the US can use this predictive modeling to better form strategies aimed at mitigating the impact of terrorism.

Solution

The Global Terrorism Database presented significant challenges due to its size and complexity, including a large number of missing (NaN) and incomplete values. To address this, we began by using MySQL to query the dataset, focusing specifically on entries related to the United States. Despite this initial step, the dataset remained unwieldy, with substantial gaps in the data. To refine the dataset further, we transitioned to Python, utilizing the Pandas library to filter and validate our dataset, even removing columns such as “approxdate”, “resolution”, “related”, and other columns that proved to be problematic and messy. Through this we successfully resolved all missing values, creating a clean and comprehensive dataset. With the data properly prepared, we were able to proceed to the next phase: selecting the appropriate variables for building our predictive model.

Despite our extensive data cleaning, we were left with 135 columns to choose from. For our predictive models, we narrowed our focus to 3–5 key variables, depending on the model. However, three variables consistently appeared in every predictive model we implemented: the year, attack type, and attacks per year. The attacks per year variable was created by further querying our dataset. We grouped the number of attacks occurring from 1970 to 2020 and visualized this variable using a bar graph:

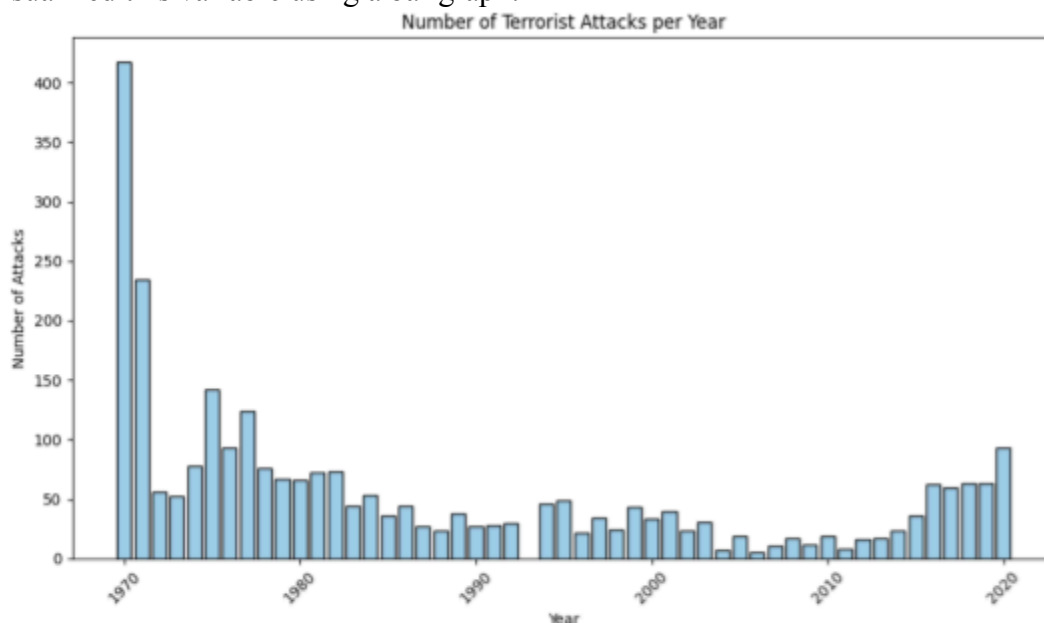


Figure 1: Number of Terrorist attacks per Year

By analyzing Figure 1, we identified a significant missing value in the data for the year 1993, which lacks recorded attacks in the Global Terrorism Database. While this gap had to be accounted for in our project, we had thought that the remaining 49 years of data would be sufficient enough for our predictive model. Another critical variable was attack type, which helped indicate the types of attacks and their frequency in previous years. Initially, this variable was represented numerically—for instance, “1” for an armed assault—which could mislead the model into interpreting it as a continuous value. To address this, we reclassified it as a

categorical variable using Python's Pandas library. The last main variable, the year, was an essential variable for our predictive models. While other variables, such as target type and additional notes, were also considered, they contributed minimal value to the model's performance and were ultimately unconsidered for our ARIMA model.

Moving on to choosing our predictive model as stated previously in the Introduction we had tried several predictive models beforehand including linear regression, poisson regression, gradient boosting, and random forest models which yielded failing results. However, the best predictive model that had worked for us was the ARIMA model. We believe that this was the best option for us because it is particularly suited to time-series data, which is the main type of what the GTD is. The ARIMA model excels at capturing time-dependent patterns, leveraging historical trends to predict future values. Since our dataset focused on "attacks per year," a clear numeric target variable with a temporal component, ARIMA was an ideal choice. Additionally, after preprocessing, the data exhibited stationarity, a critical requirement for ARIMA. This transformation enabled the model to analyze consistent trends and residual patterns effectively. Moreover, with the gap of missing data of 1993, the ARIMA's design allowed it to focus on numeric trends and handle time-series gaps. Its ability to model seasonality, trends, and residuals without overfitting made it perfect for forecasting long-term events, which is why it was our chosen model to predict future terrorist attacks for the upcoming years of 2025-2040.

Evaluation

Moving on to our evaluation of our ARIMA model we must see how the predicted values that our model generates compared to the actual values that happened through the years of 1970 to 2020. This will give us a baseline of how accurate the values of our ARIMA model is when it will predict the future values for the 2025-2040 years.

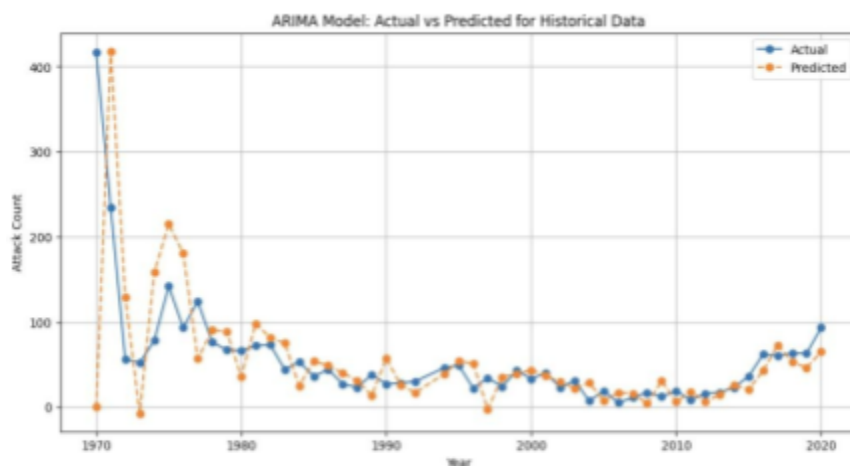


Figure 2: Actual vs Predicted Values

Figure 2, which compares actual and predicted values for historical data, highlights the ARIMA model's ability to replicate past trends with reasonable accuracy. The close alignment between the two lines indicates that the model effectively captured temporal patterns and dependencies within the dataset. However, discrepancies in certain years, such as the gap in 1993

due to missing data, highlight the limitations of the dataset rather than the model itself. When evaluating the model's performance on historical data, we calculated a percentage error of 63.36%. While this error indicates room for improvement, it is important to consider the challenges posed by missing or incomplete data and the inherent unpredictability of terrorism trends. Taking this into consideration, we will now use our ARIMA model to predict the number of terrorist attacks that could happen in the following years.

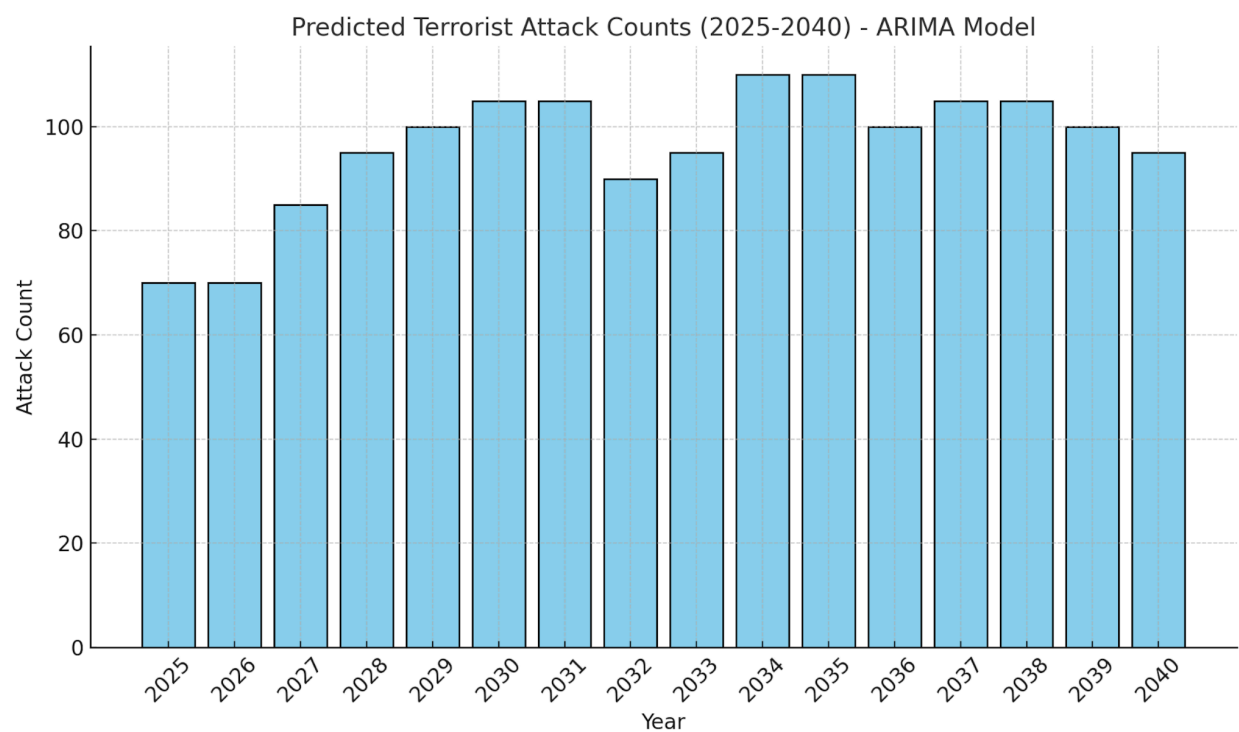


Figure 3 illustrates the predicted number of terrorist attacks for the years 2025 to 2040, showcasing the ARIMA model's ability to forecast long-term trends. The graph indicates a gradual upward trajectory, suggesting a potential increase in terrorist activity over the forecasted period. On average , the model predicts around 91 attacks per year, with values ranging from a low of 69.91 to a high of 116.38, and a standard deviation of 12.85. This projected trend aligns with historical observations and underscores the importance of ongoing analysis to anticipate and mitigate risks. While the ARIMA model successfully identified these trends, it is essential to interpret the results cautiously, considering the model’s reliance on historical data and the exclusion of external factors like political and social changes that could influence future outcomes.

Future Work/Conclusion

Overall, while our ARIMA model achieved significant insights, its prediction error of 63.36% indicates room for improvement. One limitation is the reliance on 49 years of data, which may not be sufficient for accurately forecasting long-term trends in terrorist attacks. This limitation is further compounded by the use of only three primary variables—"year," "attacks per year," and "attack type"—with only “attack type” being a categorical variable. Including

additional, more nuanced variables could enhance the model's predictive power. Future iterations of this work could explore the integration of variables that reflect broader socio-political contexts. For example, incorporating the political state of the US, its foreign policy stance, or even global geopolitical tensions could provide richer context for predictions. Furthermore, variables derived from public sentiment, such as surveys measuring perceptions of safety or trust in government, could add valuable dimensions to the model. These types of categorical variables would allow the ARIMA model to account for more complex interactions influencing terrorist activity. Despite its limitations, the ARIMA model developed in this project offers actionable insights. Its predictions, even with their inherent uncertainty, equip policymakers with data to prepare for potential increases in terrorist activity. The model's projections may appear substantial compared to recent trends, but historical patterns remind us that global events can trigger sudden and dramatic shifts in terrorism, as seen in the increase in attacks between 2019 and 2020. This underscores the importance of readiness and proactive mitigation strategies that the US should consider.

Contribution of Each Team Member

Dilan Fajardo: Contributed to querying and cleaning the dataset using Python, ensuring it was prepared for predictive modeling. Established the main variables used in the models and attempted the implementation of gradient boosting, linear regression, and Poisson regression models. Developed the "Attacks Per Year" graph to visualize trends and supported the statistical analysis of results, including calculations of percentage error, mean, minimum, mode, and standard deviation. Authored the Abstract, Solution, and Future Work/Conclusion sections of the paper,

Harish Rajagopalan: designed and implemented the Arima time series model to predict terrorist attacks in the USA, focusing on capturing historical trends. Queried the GTD to extract USA specified data, creating a csv file for further analysis and modelling. Created bar graphs for predicted attacks counts for the years 2025 to 2040, aiding in interpretation and presentation of the models forecasts. Assisted in calculating the percentage error for the actual vs predicted historical data. Authored the Introductions and Evaluations sections of the paper.

References

START (National Consortium for the Study of Terrorism and Responses to Terrorism). (2022). Global Terrorism Database 1970 - 2020 [data file]. <https://www.start.umd.edu/gtd>